



COMP 5800

Industrial Research Project
Course Coordinator: Dr Masud Karim

Proposal Report:

Security Information and Event Management

Group: C261W-5304

Name	Username	ID
Student 1	Jithu Johan Jose	A1985627
Student 2	Promise Rijal	A1980984
Student 3	Aakritansh Modgil	A3182984
Student 4	Junaid Abrar Razeen	A1991907
Student 5	Ruichao Li	A1942645
Student 6	Humayra Ashrafi	A3037578

Client/Sponsor	Eagle
Mentor	David Menzies

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— C O N T E N T S —

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SECURITY INFORMATION AND EVENT MANAGEMENT

I. EXECUTIVE SUMMARY

This proposal provides Bedminster Health management with a comprehensive understanding of the existing monitoring gap with regards to cybersecurity, the options available to bridge this gap, and the trade-offs involved with the various options of SIEM technology. By the end of this project, the management of Bedminster Health will be able to make informed decisions regarding the options available with regards to SIEM technology and the future direction that the project should take.

Although Bedminster Health generates security events across its network, the lack of analysis or correlation of this data in real-time means that any malicious activity within the network is only detected after the fact. This exposes the healthcare facility to long-lasting cyber threats, the possibility of data breaches, and non-compliance with regulations. This is a major gap in the cybersecurity system that the CISO of the healthcare facility has identified as a priority area that needs to be addressed.

The project will be completed within a period of 10 weeks with all the agreed deliverables handed over to the CISO and IT leadership department.

II. PROJECT BACKGROUND AND DESCRIPTION

Bedminster Health is a large-scale healthcare organisation with operations in one main hospital, four satellite hospitals, and several clinics. With over 6,000 staff members, 150 servers, and 2,000 workstations connected with a centralized network, the organisation is dealing with highly sensitive data and is heavily regulated in Australia. Bedminster Health has zero ability to identify any cybersecurity threats in real time.

They are generating massive amounts of security event data every single day on their five hospital sites, 150 servers, and 2,000 workstations. However, they currently have no ability to analyse the data which means that any threats are only discovered after a major breach or security event has occurred. This leaves them vulnerable to data breaches, ransomware threats, and legal action against them because of the strict regulations in Australia. A SIEM solution would help their organization to centralize a system where they could track all the logs for forensics. For Bedminster Health, any prolonged outage or breach could affect patient care, disrupt hospital operations and damage community trust. The CISO's request for a SIEM business case reflects a strategic decision to move from reactive incident response to consistent monitoring and early detection ([SecureOps, 2025](#)).

The purpose of this project is to fulfil this need by researching and evaluating the leading available SIEM solutions on the market and providing a justified recommendation to the leadership of Bedminster Health. The solution will be analysed in terms of the organization's technical capabilities, personnel availability, regulatory requirements and financial constraints. The end goal of this project is to create a business case that will enable the leadership to make an informed decision regarding the implementation of a SIEM solution.

III. PROJECT SCOPE

This project will encompass the entire range of work involved in producing a business case for implementation, from the initial identification of the organisation's current security monitoring needs to the development of a compliant and costed solution architecture. The actual implementation of the proposed solution will not form part of this project and will instead be considered a separate project once approval has been forthcoming from the executive.

a. INCLUSIONS

The following aspects of the project are included in the scope:

- Review of contemporary security monitoring practices in large organisations and difficulties being experienced..
- Overview of the main types of systems and devices in use and how they would need to interact with a future SIEM solution at a high level.
- Comparison of several suitable SIEM options against Bedminster Health's size, resources, budget, and regulatory obligations, leading to a recommended option.
- High-level description of how the recommended SIEM solution could operate across the organisation's different sites and environments.
- Clear grouping of different system and device types to show how each group would be handled within the proposed solution.
- Outline of how security information could be stored and organised over time to balance day-to-day monitoring needs with longer-term record-keeping.
- High-level approach for bringing past security information into the new solution in a way that supports future analysis and reporting.
- Summary of how the recommended approach supports Bedminster Health's key Australian cybersecurity, privacy, and critical infrastructure obligations.
- Indicative cost model showing estimated ongoing costs and how these relate to the organisation's scale and usage.
- High-level staged roadmap showing how the recommended solution could be rolled out over time using the existing IT team.

b. EXCLUSIONS

The following aspects of the project are explicitly out of scope:

- Implementation or configuration of any SIEM system in Bedminster Health's live environment.
- Ongoing, real-time security monitoring or incident response activities after a SIEM is implemented.
- Security monitoring or management of medically certified clinical devices, which would require a separate specialised project.
- Contract negotiation, licence purchasing, or any formal commercial agreements with vendors.
- Design and delivery of staff training or security awareness programs related to the SIEM.
- Changes to existing hospital policies, procedures, or governance frameworks beyond high-level recommendations in the business case.

These inclusions and exclusions are informed by preliminary research into common SIEM implementation challenges in healthcare and align with the scope defined in the Bedminster Health project brief.

IV. RISK

This risk register identifies the key risks associated with the Bedminster Health SIEM project, covering team coordination, workload distribution, scope management, technical knowledge gaps, tool access, and information limitations. Each risk has been assessed by probability and impact, with mitigation strategies and residual risks documented to support proactive management throughout the project lifecycle.

Risk	Probability (out of 1.0)	Impact	Mitigation Strategy
Team scheduling conflicts affecting meetings and deadlines	0.6	High	To ensure all team members remain informed and aligned, maintain a shared schedule accessible to everyone. Conduct meetings via online platforms to maximize participation, and distribute detailed meeting notes promptly so that absent members can stay up to date. Sessions should be recorded and made available for those who were unable to attend, allowing them to review discussions and get back on track at their convenience.
Uneven workload across team members leading to delays	0.5	High	Review task allocation weekly to rebalance tasks and use the task board to track ownership clearly.
Ambiguous or changing project scope causing rework	0.4	High	Document scope in the Project Charter and make sure to have team agreement before accepting scope changes.
Limited experience with SIEM and cybersecurity concepts slowing progress	0.5	Medium	Allocate extra time in early research phases and do share summary notes to pair less experienced members with Research/Technical Lead.

Difficulty accessing or using required tools	0.3	Medium	Test tools early and have simple backups (Excel, email) but raise access issues quickly with support.
Incomplete or unclear information about Bedminster Health's environment impacting assumptions	0.4	Medium	Clearly document all assumptions and flag uncertainties in the report to validate assumptions with the lecturer where possible.

V. BUDGET

This budget provides an overview of the projected costs involved in completing the project, accounting for team labour, necessary tools, and any physical submission requirements. Costs have been kept to a minimum by leveraging freely available and university-licensed resources, with the most significant investment being the collective time and effort contributed by the team throughout the project.

Budget Item	Estimated Cost	Justification
Project tools and resources	\$0.00	All tools used to produce this report are available at no cost via free tiers or university student licensing, including Microsoft 365, collaboration platforms, and diagramming tools.
Student Labor — Presentation Preparation	\$2,700.00	Covers time allocated to slide deck development, speaking role preparation, and rehearsal sessions in the final project phase.
Student labor research and report writing 6 members x 150hrs (15 hrs. per week) @ AUD \$45/hr.	\$40,500.00	Reflects time invested across all team members in literature review, architecture design, Azure cost modelling, and report drafting. Rate based on approximate junior IT analyst casual wage in South Australia
Printing & Binding — Final Submission	\$48.00	Covers color printing of the full project report including charts, diagrams, and appendices, plus professional binding for each team member's submission copy.
Travel Expenses	\$700.00	Covers public transport or fuel reimbursement for team members attending scheduled in-person collaboration and review sessions at university campus.
Contingency Reserve (5%)	\$2,184.40	Allocated to cover unforeseen expenses such as additional printing, unexpected travel, or supplementary resource requirements arising during the project.
Meeting Room / Collaborative Space Hire	\$225.00	Cost of booking a university study room or co-working space for structured team workshops, design sessions, and review meetings where a shared physical environment is required.
Total Project Cost	\$46,357.40	

VI. ROLES AND RESPONSIBILITIES

This section outlines the roles and responsibilities assigned to each team member to ensure clear accountability and effective collaboration throughout the project. Responsibilities have been distributed to align with individual strengths, covering technical, communication, documentation, and coordination functions, alongside academic mentorship and client guidance.

Team Member	Role	Key Responsibilities
Promise Rijal	Technical Lead	Lead SIEM solution compatibility and cost estimation gather requirements for the report.
Aakritansh Modgil	Communication Lead	Coordinate and manage all internal team communications, meeting schedules, and progress updates to ensure alignment across all project phases.
Jithu Johan Jose	Task Manager	Coordinate team activities, maintain the Task Planner, track milestones, and ensure deliverables are completed on schedule.
Humayra Ashrafi	Documentation Lead	Consolidate all team contributions into cohesive report sections and maintain consistent formatting and writing standards.
Ruichao Li	Records Lead	Maintain the team time log, and record and distribute meeting minutes.
Junaid Abrar Razeen	Supporting Researcher and Reviewer	Conduct critical research on SIEM capabilities tailored to healthcare environments, and critically evaluate all drafted sections to ensure the proposed architecture meets the specific security and compliance needs of Bedminster Health.
David Menzies	Project Mentor	Provides academic guidance, reviews progress at milestones, and evaluates deliverables against assessment criteria.
Wenhao Liang	Client	Defines project requirements, clarifies scope and evaluates the final solution against agreed evaluation criteria.

i. COMMUNICATION PLAN

This communication plan outlines how information will be shared among the project team, client, and mentor throughout the project lifecycle. Regular and structured communication channels have been established to ensure all stakeholders remain informed, aligned, and able to provide timely input across each phase of the project.

Stakeholder	Communication Method	Frequency	Purpose
Project Team	Microsoft Teams & Whatsapp	Two times a week	Discuss project progress, assign tasks and resolve issues

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Client	Email and in person	Weekly	Provide project updates and clarify project requirements
Project Mentor	Email and in person	Weekly	Seek academic guidance and feedback on project direction

ii. PROJECT STAKEHOLDER

This section identifies the key stakeholders involved in or affected by the Bedminster Health SIEM project, along with their respective roles and interests. Understanding each stakeholder's perspective ensures the project team can address their needs and expectations throughout the proposal development process.

Name	Role	Interest in the project
David Menzies	Project Mentor	Ensures the project meets learning outcomes, provides guidance on methodology, and evaluates deliverable quality against assessment criteria
Wenhao Liang	Client	Wants a realistic, well-justified SIEM proposal that addresses the organization's cybersecurity gaps within the project brief constraints
Chief Information Security Officer (CISO)	Cybersecurity policy owner	Concerned with whether Sentinel meets compliance obligations, reduces organizational risk, and aligns with existing security policies
IT Security Team	Primary end users of Microsoft Sentinel	Need a platform that is manageable with a 30-person team, produces actionable alerts, and integrates with existing infrastructure across all 5 sites
IT System Administrators	Responsible for agent deployment and infrastructure	Concerned with the technical complexity of deploying AMA agents across 150 servers and 2,000 workstations without disrupting existing systems
Hospital Administration / Executive Leadership	Budget approvers and operational decision makers	Interested in cost-effectiveness, return on investment, regulatory compliance, and minimal disruption to hospital operations
Clinical Staff	Indirect end users - patient care providers	Need assurance that the SIEM deployment will not interrupt access to clinical systems, electronic health records, or medical devices
Project Team	Project deliverable owners	Responsible for producing a thorough, well-researched proposal that accurately represents Bedminster Health's needs and recommends a defensible solution

VII. DELIVERABLES & PROJECT EVALUATION CRITERIA

The following criteria will be used to evaluate the success of this project from the client's perspective. These criteria have been agreed upon by all project parties and will serve as the primary basis for assessing the completeness and appropriateness of all final deliverables.

Evaluation Criteria	Description	Measure of Success
A business case is developed	The business case is a document outlining a recommended solution and its implementation plan	The proposed solution must be financially viable and clearly justified for a mid-sized healthcare organisation
Scope coverage	Inclusions are defined in the project proposal	All in scope activities and matters have been addressed
Risk coverage	Documentation of the risks associated with the recommended solution	All identified project and implementation risks must be documented with clear mitigation strategies with assigned ownership and clear acceptance criteria
Documentation quality	All deliverables must be professionally written, well-structured, and immediately actionable by the client	The report is clearly written, fully cited, free of unsupported assumptions, and all architecture decisions are explicitly justified

VIII. PROJECT IMPLEMENTATION

The implementation of this project follows a structured yet dynamic sequence, where phases overlap and run in parallel to maximise efficiency. Each phase has defined objectives and outputs that feed into subsequent activities, ensuring research, analysis, and documentation build progressively toward the final deliverable.

1. PHASE 1 – PROJECT INITIATION

- Establish roles and responsibilities across the project team.
- Define project scope and confirm key deliverables.
- Identify resource requirements and finalize the project budget.
- Progressively develop and complete the Project Charter by the proposal deadline of 20 March 2026.

2. PHASE 2 – RESEARCH SPRINT

- Commence three concurrent workstreams immediately after the proposal submission.
- Workstream A: Map all devices, sites, and log sources across Bedminster Health's five locations.

- Workstream B: Identify functional and non-functional requirements and validate assumptions with the client and mentor.
- Workstream C: Research SIEM solutions in healthcare environments and review Australian regulatory obligations including the Essential Eight and the Privacy Act.

3. VENDOR QUOTE OUTREACH

- Contact all shortlisted SIEM providers to request formal pricing quotes.
- Actively follow up with providers to chase outstanding responses.
- Compile all received quotes into a structured log for use in the cost evaluation stage.

4. PHASE 3 – EVALUATION & DESIGN

- Conduct a structured comparison of all shortlisted SIEM platforms against defined requirements.
- Perform vendor cost analysis using compiled quotes from the outreach track.
- Select the most suitable SIEM solution and document the recommendation with clear justification.
- Produce a high-level architecture design showing how the solution will operate across all five sites.

5. PHASE 4 - STRATEGY & DEPLOYMENT PLANNING

- Develop a phased rollout model and deployment sequencing across all sites.
- Plan AMA agent deployment across servers and workstations.
- Design the incident response and alert handling workflow.
- Produce detailed cost modelling and budget justification for the final business case.

6. PHASE 5 – CONSOLIDATION BUFFER & INTERNAL REVIEW

- Conduct peer review of all drafted sections.
- Cross-check assumptions and flag any remaining gaps or uncertainties.
- Hold a mentor check-in to confirm alignment with assessment expectations before report writing begins.

7. PHASE 6 – REPORT WRITING & DOCUMENTATION

- Integrate all phase outputs into a unified final report structure.
- Apply consistent formatting and referencing standards throughout.
- Conduct a final peer review and obtain team sign-off before submission preparation.

8. PHASE 7 – PRESENTATION & SUBMISSION

- Develop the presentation slide deck and assign speaking roles across the team.
- Rehearse and refine the presentation based on internal feedback.
- Complete the final submission by the deadline of 22 May 2026.

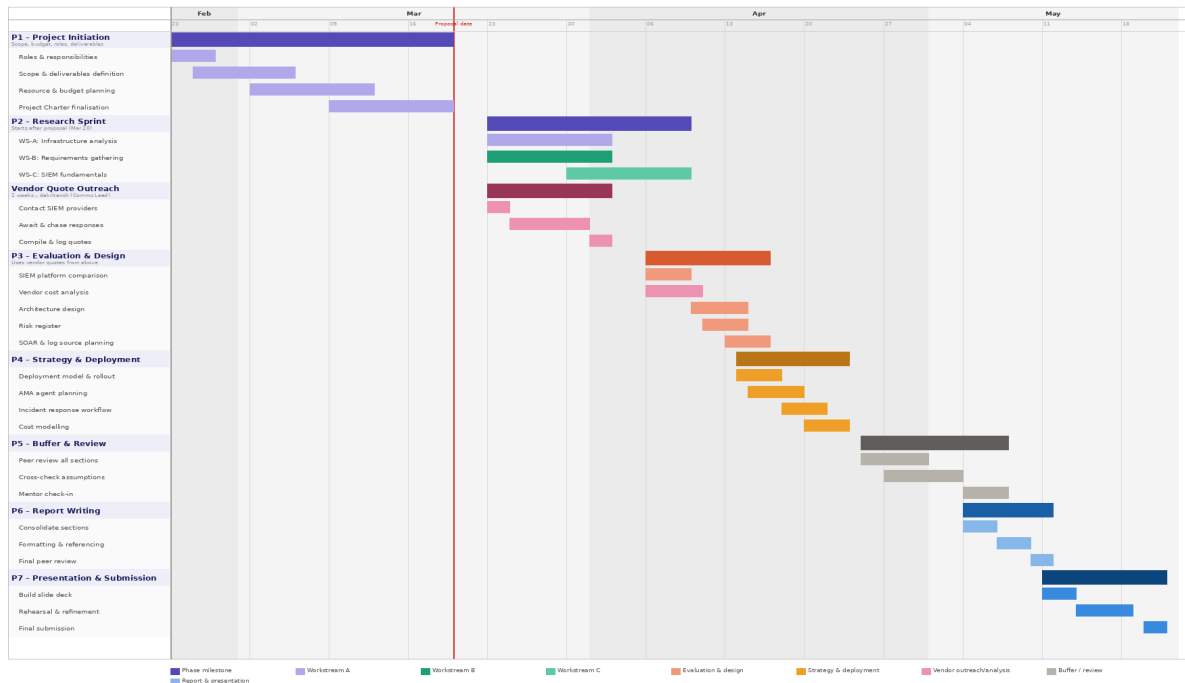
IX. PROJECT SCHEDULES

Bedminster Health SIEM Project - Implementation Plan

#	Task Name	Duration	Start	Finish	Assigned To	Predecessors
1	Phase 1 - Project Initiation	26 days	29/02/26	26/03/26	Full Team	
	Roles & responsibilities	5 days	29/02/26	27/02/26	Full Team	
	Scope & deliverables definition	10 days	29/02/26	06/03/26	Full Team	
	Resource & budget planning	12 days	02/03/26	13/03/26	Full Team	
	Project Charter finalisation	12 days	09/03/26	20/03/26	Full Team	
2	Phase 2 - Research Sprint	19 days	29/03/26	18/04/26	Full Team	1
	WS-A: Infrastructure analysis	12 days	29/03/26	09/04/26	Technical Lead	
	WS-B: Requirements gathering	12 days	29/03/26	09/04/26	Documentation Lead	
	WS-C: SIEM fundamentals research	12 days	29/03/26	19/04/26	Researcher / Reviewer	
3	Vendor Quote Outreach (Parallel)	12 days	29/03/26	09/04/26	Communications Lead	1
	Contact SIEM providers	3 days	29/03/26	29/03/26	Communications Lead	
	Await & chase responses	7 days	29/03/26	05/04/26	Communications Lead	
	Compile & log quotes	3 days	02/04/26	03/04/26	Communications Lead	
4	Phase 3 - Evaluation & Design	12 days	06/04/26	17/04/26	Technical Lead	2, 3
	SIEM platform comparison	5 days	06/04/26	10/04/26	Technical Lead	
	Vendor cost analysis	6 days	06/04/26	12/04/26	Technical Lead	
	Architecture design	6 days	10/04/26	15/04/26	Technical Lead	
	Risk register	5 days	11/04/26	15/04/26	Task Manager	
	SOAR & log source planning	5 days	13/04/26	17/04/26	Technical Lead	
5	Phase 4 - Strategy & Deployment	11 days	14/04/26	24/04/26	Technical Lead	4
	Deployment model & rollout	5 days	14/04/26	18/04/26	Technical Lead	
	AMA agent planning	6 days	15/04/26	20/04/26	Technical Lead	
	Incident response workflow	5 days	18/04/26	22/04/26	Researcher / Reviewer	
	Cost modelling	5 days	20/04/26	24/04/26	Documentation Lead	
6	Phase 5 - Buffer & Internal Review	14 days	25/04/26	08/05/26	Full Team	5
	Peer review all sections	7 days	25/04/26	01/05/26	Researcher / Reviewer	
	Cross-check assumptions	8 days	27/04/26	04/05/26	Full Team	
	Mentor check-in	5 days	04/05/26	08/05/26	Task Manager	
7	Phase 6 - Report Writing	9 days	04/05/26	12/05/26	Documentation Lead	6
	Consolidate sections	4 days	04/05/26	07/05/26	Documentation Lead	
	Formatting & referencing	4 days	07/05/26	10/05/26	Documentation Lead	
	Final peer review	3 days	10/05/26	12/05/26	Researcher / Reviewer	
8	Phase 7 - Presentation & Submission	12 days	11/05/26	22/05/26	Full Team	7
	Build slide deck	4 days	11/05/26	14/05/26	Communications Lead	
	Rehearsal & refinement	6 days	14/05/26	19/05/26	Full Team	
	Final submission	3 days	20/05/26	22/05/26	Task Manager	

Bedminster Health SIEM Project - Implementation Gantt Chart

All research phases commence after Business Proposal submission on 20 March 2026. Vendor quote outreach: 2 weeks (Mar 23 – Apr 5), assigned to Communications (Comm. Lead).



The Gantt chart outlines eight project phases and concurrent workstreams running from 23 February to 22 May 2026. A proposal deadline marker on 20 March 2026 separates the initiation period from all subsequent research and technical phases.

X. APPROVAL

The undersigned members of project group **C261W-5304** hereby confirm that they have reviewed the contents of this project proposal and agree that it accurately represents the planned approach, scope, and deliverables for the Industrial Research Project. By signing below, each party acknowledges their role and responsibilities as outlined in this document and provides formal approval for the project to proceed in accordance with the terms described herein.

Name	Role	Signature	Date
Aakritansh Modgil	Student		20/03/2026
Humayra Ashrafi	Student		20/03/2026
Jithu Johan Jose	Student		20/03/2026
Junaaid Abrar Razeen	Student		20/03/2026
Promise Rijal	Student		20/03/2026
Ruichao Li	Student	Ruichao Li.	20/03/2026
David Menzies	Mentor		20/03/2026

XI. REFERENCES

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